

# Diagnosing Separation and Initial-Condition Errors

## Answer Key

### AP Calculus AB/BC • Error Analysis

#### Quick Answers

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|------------------|-----------|---------|
| 1. $2x^4 - 3x^2$ | 4. 890.48 | 7. 6.83 |
| 2. 25            | 5. 5      | 8. —    |
| 3. 8.15          | 6. 4      |         |
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#### Common wrong answers

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- If they got 20: dropped the constant of integration before evaluating the particular solution*
  - If they got 4: integrated the right side as though  $y$  were a constant, instead of separating the variables first*
  - If they got 153.41: added the constant after exponentiating instead of carrying it through as a multiplying factor*
  - If they got 7: took the square root of each term separately, turning a sum of squares into a sum*
  - If they got 1.33: lost the minus sign from the integral of  $y$  to the negative second power*
  - If they got 4.83: left the constant of integration out, reporting the antiderivative rather than the particular solution*
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**1. General solution of  $\frac{dy}{dx} = 8x^3 - 6x$ .**

**Step 1 (check for separation).** The right side holds no  $y$ , so the variables are already separated and the equation integrates directly. **Step 2 (integrate).**  $y = \int (8x^3 - 6x) dx$ ,

so

$$y = 2x^4 - 3x^2 + C$$

. **Step 3 (interpret  $C$ ).** On a slope field,  $C$  selects which member of the family of parallel solution curves you are on — every curve has the same slope at a given  $x$ , so they differ only by a vertical shift.

**2. Same equation with  $y(1) = 4$ ; evaluate  $y(2)$ .**

**Step 1 (integrate first).** The general solution is  $y = 2x^4 - 3x^2 + C$ . The initial condition can be used only now, once a  $C$  exists to solve for — that is the step it belongs to. **Step 2 (apply the condition).**  $y(1) = 2 - 3 + C = 4$ , so  $C = 5$  and the particular solution is  $y = 2x^4 - 3x^2 + 5$ . **Step 3 (evaluate).**  $y(2) = 32 - 12 + 5 = \boxed{25}$ . A student who integrates and then evaluates without solving for  $C$  reports 20 — the antiderivative, not the solution of this initial-value problem.

**3. Find and fix:**  $\frac{dy}{dx} = 2xy$ ,  $y(0) = 3$ , **find**  $y(1)$ .

**Step 1 (name the first wrong line).** The very first line. Wren writes  $\int dy = \int 2xy dx$  and then integrates the right side as if  $y$  were a constant. It is not:  $y$  is the unknown *function* of  $x$ , so  $\int 2xy dx$  cannot be evaluated at all until the  $y$  is moved to the other side.

**Step 2 (separate, then integrate).**  $\frac{dy}{y} = 2x dx$ , so  $\ln|y| = x^2 + C$  and  $y = Ae^{x^2}$  with  $A = e^C$ . **Step 3 (apply the condition and evaluate).**  $y(0) = A = 3$ , so  $y = 3e^{x^2}$  and  $y(1) = 3e \approx \boxed{8.15}$ . Check:  $y' = 3e^{x^2}(2x) = 2xy \checkmark$ . Wren's 4 is the answer to a different, non-separable computation.

**4. Find and fix:**  $\frac{dy}{dx} = 5y$ ,  $y(0) = 6$ , **find**  $y(1)$ .

**Step 1 (name the first wrong line).** The second.  $\ln|y| = 5x + C$  is correct; exponentiating it is not  $y = e^{5x} + C$ . Exponentiating a *sum* gives a *product*:  $e^{5x+C} = e^C e^{5x}$ . **Step 2 (carry the constant as a factor).** Write  $A = e^C$ , so the general solution is  $y = Ae^{5x}$  — the constant multiplies, it does not add. **Step 3 (apply the condition and evaluate).**  $y(0) = A = 6$ , so  $y = 6e^{5x}$  and  $y(1) = 6e^5 \approx \boxed{890.48}$ . Check:  $y' = 30e^{5x} = 5y \checkmark$ . Dev's 153.41 fails that check, which is the one-line test that would have caught the error before any arithmetic.

**5. Find and fix:**  $\frac{dy}{dx} = \frac{x}{y}$ ,  $y(0) = 4$ , **find**  $y(3)$ .

**Step 1 (name what is wrong).** The square root was taken term by term.  $\sqrt{u^2 + v^2}$  is not  $u + v$ ; the radical must be applied to the whole side at once, after the constant has been folded in. **Step 2 (do it correctly).** From  $\frac{y^2}{2} = \frac{x^2}{2} + C$ , multiply through by 2 to get  $y^2 = x^2 + K$  with  $K = 2C$ . Applying  $y(0) = 4$  gives  $16 = 0 + K$ , so  $K = 16$  and  $y = \sqrt{x^2 + 16}$  (the positive root, because  $y(0)$  is positive). **Step 3 (evaluate).**  $y(3) = \sqrt{9 + 16} = \boxed{5}$ . Check:  $y' = \frac{x}{\sqrt{x^2 + 16}} = \frac{x}{y} \checkmark$ . Rosa's 7 is exactly  $3 + 4$ , the signature of the term-by-term root.

**6. Find and fix:**  $\frac{dy}{dx} = y^2$ ,  $y(0) = 2$ , **find**  $y(0.25)$ .

**Step 1 (name the missing sign).**  $\int y^{-2} dy = -\frac{1}{y}$ , not  $+\frac{1}{y}$ . The power rule sends  $y^{-2}$  to  $\frac{y^{-1}}{-1}$ , and Sam dropped that minus. **Step 2 (redo with the sign).**  $-\frac{1}{y} = x + C$ , and  $y(0) = 2$  gives  $-0.5 = C$ . So  $-\frac{1}{y} = x - 0.5$ , that is  $y = \frac{1}{0.5 - x}$ . **Step 3 (evaluate and read the behaviour).**  $y(0.25) = \frac{1}{0.25} = \boxed{4}$ . As  $x \rightarrow 0.5^-$  the denominator goes to zero and  $y \rightarrow \infty$ : the solution blows up in finite time, so its interval of existence stops at  $x = 0.5$ . Sam's version,  $y = \frac{1}{x + 0.5}$ , decays smoothly forever and hides that blow-up entirely — the sign error changes the qualitative behaviour, not just a digit.

7.  $\frac{dy}{dx} = \frac{6x}{x^2 + 1}$ ,  $y(0) = 2$ ; **evaluate**  $y(2)$ .

**Step 1 (integrate).** With  $u = x^2 + 1$  the right side integrates to  $3\ln(x^2 + 1)$ , so  $y = 3\ln(x^2 + 1) + C$ . **Step 2 (apply the condition).**  $y(0) = 3\ln 1 + C = C = 2$ , so  $y = 3\ln(x^2 + 1) + 2$ . Applying the condition *before* integrating would be meaningless: there is no constant to determine yet, and  $\frac{dy}{dx}$  is a statement about a slope, not about a value of  $y$ .

**Step 3 (evaluate).**  $y(2) = 3\ln 5 + 2 \approx \boxed{6.83}$ . A student who never writes the constant reports 4.83 — the antiderivative evaluated at 2, which satisfies the differential equation but not the initial condition.

8. **Refute:** “a constant may be added before or after exponentiating.” *Open reasoning — read by hand.*

**Model argument.** Differentiate the claimed solution: if  $y = e^{5x} + 5$  then  $y' = 5e^{5x}$ , while  $5y = 5e^{5x} + 25$ . These differ by 25 for every  $x$ , so the claimed function does not satisfy  $\frac{dy}{dx} = 5y$  and the placement of the constant is not a matter of taste. **What the constant becomes.** Integrating gives  $\ln|y| = 5x + C$ ; exponentiating turns the sum into a product,  $y = e^C e^{5x} = Ae^{5x}$ . So the additive constant of integration becomes a *multiplicative* factor  $A$ , and  $A$ , not  $C$ , is what the initial condition determines. **Accept:** any argument that produces the mismatch by differentiation, or that shows  $y = e^{5x} + 5$  fails the equation at a single specific  $x$  (for example  $x = 0$ :  $y' = 5$  but  $5y = 30$ ). **Do not accept:** “because the book says  $Ae^{5x}$ ” with no test, or a claim that both forms work on some interval.